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Evaluating the Relationship Between Emotion Regulation, Mood and Distressing Dreams Using Daily Sleep Diary Reports in Trauma Survivors

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ABSTRACT

Posttraumatic stress disorder (PTSD) is a debilitating psychiatric condition that affects approximately 6% of the US population, with even higher rates among veterans. Sleep disturbances (e.g., distressing dreams) are among the most reported symptoms in individuals with PTSD. Distressing dreams (sometimes referred to as nightmares) have been associated with significant negative mental health outcomes, such as suicide. Therefore, understanding factors that contribute to or mitigate the risk of experiencing distressing dreams is critical for improving mental health in people with PTSD. The current study uses data on twice daily mood ratings from a mobile phone application across 3 weeks of follow-up to assess the bidirectional relationship between mood and distressing dreams. Additionally, we investigate how emotion regulation difficulties contribute to the incidence of distressing dreams and to their impact on overall mood. Participants were veterans ($N=90$) and civilians ($N=13$) enrolled as part of a larger study of distressing dream physiology in trauma-exposed individuals. Our results suggest that among individuals with poor mood before bed, the risk of having at least one distressing dream increases, resulting in poor mood the following morning. This negative mood state carried over into the evening on the next day, thereby further increasing the risk of having another distressing dream. Adjusting for sex, age and PTSD severity, we find that individuals with better emotion regulation have a lower incidence of distressing dreams and better mood on average. Future research on novel interventions targeting emotion regulation, mood and/or sleep disturbances is warranted.

1 | Introduction

Posttraumatic stress disorder (PTSD) is present in roughly 6%–8% of the US population (Kessler et al. 2008). In military populations, PTSD affects approximately 14% of veterans (Tanielian et al. 2008) and is one of the most prevalent mental disorders experienced by US veterans (Wisco et al. 2022; Richardson et al. 2010). Among veterans (Seal et al. 2007;

Richards et al. 2020) and trauma-exposed civilians (Milanak et al. 2019), sleep disturbances are commonly reported. A large study with Operation Enduring Freedom (OEF)/Operation Iraqi Freedom (OIF) veterans ($N=103,788$) found that approximately 70% of participants with PTSD experienced sleep disturbances (Seal et al. 2007). Similarly, in a national sample of US adults ($N=2647$) who were trauma-exposed, 277 individuals met the Diagnostic and Statistical Manual of Mental

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Disorders Fifth Edition (DSM-5; APA 2013) criteria for PTSD and, of those, 92% reported sleep disturbances (Milanak et al. 2019).

Among civilians and veterans, distressing dreams are considered a hallmark of PTSD (APA 2013; Weathers et al. 2013; Ross et al. 1989) and a core feature of trauma-related sleep disturbances (i.e., difficulties with sleep that arise following exposure to a traumatic event; Weber and Wetter 2022). Of note, while many researchers use the term 'nightmare' when reporting on distressing dream experiences in trauma subjects, definition of the term 'nightmare' differs widely across studies, and the term 'distressing dream' captures the core phenomenon investigated across studies. This core phenomenon involves the experience of an unpleasant dream which is recalled upon awakening (whether or not the dream itself awakens the dreamer), and which causes unpleasant emotions, typically fear or anxiety. The term 'nightmare' is often, but not always, used to refer to a particularly distressing dream that awakens the dreamer. Notwithstanding differences in definitions or operationalization across studies, taken together, the evidence underscores that distressing dreams are a key feature of PTSD and are associated with serious negative mental health outcomes. For example, in a secondary analysis of data from a treatment study for nightmares that compared nightmares in trauma-exposed civilians with lifelong ($N=13$; 24.1%) versus post-trauma nightmares ($N=41$; 75.9%) the authors found that when the onset of nightmares began after a traumatic experience, individuals endorsed greater depression, posttraumatic symptoms and poorer sleep quality compared to individuals who experienced lifelong nightmares (Davis et al. 2011). This highlights the clinical relevance of addressing mental health outcomes associated with trauma-related distressing dreams among civilians. Moreover, in a study by Creamer et al. (2018) examining the prevalence of distressing dreams and the clinical characteristics of distressing dream sufferers in a sample of 500 United States military personnel, the authors reported that 31.2% experienced weekly distressing dreams (referred to as 'bad' dreams based on the Pittsburgh Sleep Quality Index [PSQI; Buysse et al. 1989]), and that endorsing weekly distressing dreams was associated with a higher likelihood of an insomnia, anxiety, depression or PTSD diagnosis. Furthermore, experiencing both weekly distressing dreams as well as disruption of sleep by 'memories or nightmares of a traumatic experience,' based on the PSQI, was associated with a higher likelihood of insomnia, PTSD, depression and anxiety diagnosis, compared to experiencing frequent distressing dreams alone. With the caveat that distressing dreams could not be distinguished from sleep-disrupting memories in this report, the study nonetheless demonstrates that sleep-disrupting trauma-related mentation occurred in 60% of subjects with distressing dreams. These trauma-related sleep events were associated with worse mental health outcomes (Creamer et al. 2018). A recent systematic review underscores the association between distressing dreams (referred to as nightmares but variably defined across studies) and negative mental health outcomes (Sheaves et al. 2023), including suicidal ideation and attempts in civilian (Nadorff et al. 2011; Sjöström et al. 2009) and military (Worley et al. 2025) populations.

There is less empirical data on the factors that contribute to the incidence of distressing dreams on any particular night

and how such distressing dreams impact an individual the subsequent day. In a prospective study evaluating the behavioural effects of distressing dreams—referred to as 'nightmares' and defined as dreams that frighten the dreamer and are well remembered, but that might or might not wake the individual—Köthe and Pietrowsky (2001) found that nightmares were related to negative daytime mood and higher stress before and after sleep. Moreover, Ohayon et al. (1997) examined the prevalence of distressing dreams and their relationship to psychopathology and daytime functioning among individuals with insomnia. They specifically studied nightmares, defined according to the DSM-IV nightmare disorder criteria, which involves repeated awakening from the major sleep period with detailed recall of extended and extremely frightening dreams (APA 1994). They found that nightmare sufferers complained that their sleep disorders adversely impacted their daytime functioning, cognitive impact and daytime mood. The highest endorsed mood-related symptom among those that complained about their nightmares was 'irritable following poor night's sleep,' (39%) followed by 'anxious following a poor night's sleep' (37.4%) (Ohayon et al. 1997). Collectively, these results suggest that distressing dreams negatively impact an individual's mood and overall daytime functioning. However, the cross-sectional nature of these studies limits our understanding of these relationships. A recent study was conducted with 27 US combat veterans who completed 7 days of ecological momentary assessment to assess daytime negative affect and event-related stress following trauma-related nightmares (TRN). For that study's self-report assessments, a TRN was operationalised as a distressing dream that was at least slightly vivid and had some similarity to the memory of the traumatic event (based on a score >0 on scales of vividness and similarity). In their analysis, the authors focused specifically on TRN that woke participants from sleep. They found that negative affect and event-related stress were not associated with TRN the subsequent night, but that a TRN was associated with greater next day negative affect and stress (Miller et al. 2024). This suggests that the experience of a TRN may carry over into the following day. Now, research with larger samples and a longer period of data collection is sorely needed. Additionally, research examining distressing dreams defined more broadly, is critical to further examine these relationships, including Miller and colleagues' counterintuitive finding that a negative affective state did not increase the likelihood of subsequent evening TRN.

Beyond examining relationships between single distressing dream events and preceding and subsequent mood states, the role of emotion regulation tendencies may be related to distressing dreams and their consequences. Nielsen and Levin (2007) proposed a neurocognitive model of dreaming and suggested that nightmares are evidence of failures in emotion regulation. Further, they note that nightmares occur as a result of a state factor, 'affect load' (ongoing accumulation of emotional negative events that influence one's capacity to effectively regulate emotion) and a trait-like factor, 'affect distress' (dispositional tendency to experience heightened distress in response to emotional stimuli) (Nielsen and Levin 2007; Levin and Nielsen 2009). In a study investigating this model, Levin and Nielsen (2009) found that affect load accounted for all the unique variance in predicting distressing dream (i.e., nightmare and bad dream) frequency and affect distress accounted for all the unique

variance in predicting disturbed dreaming distress, a key factor in predicting psychopathology.

In line with neurocognitive nightmare models, trait-like features such as emotion regulation may be important to consider as a key contributor to distressing dreams, including nightmares, and their next-day sequelae. A recent study found an association between difficulties with affect regulation and nightmares, as well as suicide attempts, among psychiatric inpatient samples ($N=2683$) (Rufino et al. 2020). Similarly, in a study with adults admitted to an inpatient psychiatric hospital ($N=1215$), moderation analyses revealed that the association between distressing dream/nightmare severity (Disturbing Dream and Nightmare Severity Index) and depression symptoms (Patient Health Questionnaire), and distressing dream/nightmare severity and suicide risk (based on the Suicide Behaviours Questionnaire) was strongest among those who had experienced greater difficulties with emotion regulation (Blanchard et al. 2024). That said, these studies used self-report data collected at a single time point which, in turn, limit the granularity of the data and ability to examine effects of trait emotion regulation on day-to-day relationships between nightmares and mood states.

While research has been done concerning factors that contribute to nightmares, more research is needed using temporally precise data to better understand these factors. As such, the current study obtained app-based twice-daily mood ratings to assess the bidirectional relationship between negative affect states and distressing dreams among trauma-exposed individuals. Additionally, we investigated how difficulties with affect regulation contributed to the incidence of distressing dreams and overall mood. Given what is known about the gaps in the current literature, we specifically tested the following hypotheses: (1) mood in the morning predicts mood in the evening (H1a), which in turn (2) predicts the risk for having one or more distressing dreams (H1b). We hypothesised distressing dreams would then predict mood on the following morning (H1c), leading to feedback loops between worsening mood states and increased risk for distressing dreams. To determine how between-subject differences in emotion regulation difficulties impact distressing dream burden and mood, we predicted that (1) distressing dream burden and negative mood would be correlated (H2a) and that (2) difficulties with emotion regulation at baseline would predict both distressing dream burden and mood (H2b). Finally, we also predicted that participants with poor emotion regulation skills would have an increased risk of experiencing a distressing dream following worse mood in the evening (H3a) and that morning mood would be worse following a distressing dream for participants with high difficulties with emotion regulation (H3b).

2 | Methods

2.1 | Participants

Potential participants were veterans and civilians recruited nationally via clinician referrals and online advertising (e.g., using social media postings, facilitated by research recruitment services). Participants in the current study were enrolled as part of a larger study of distressing dream physiology among trauma-exposed individuals. Eligible participants had a history of trauma based on

the Trauma History Questionnaire (THQ; Hooper et al. 2011). To recruit participants with frequent, severe distressing dreams, eligible participants also had one or more distressing dreams that qualified as a nightmare, according to the DSM-5 nightmare disorder definition (APA 2013). In the initial eligibility screening for the study, a nightmare was specifically defined as a highly distressing dream that awakens the participant and is well remembered. Following initial screening, the distressing dream criterion was then assessed using a modified version of the Pittsburgh Sleep Quality Index Addendum (PSQI-A; Germain et al. 2005) to confirm that the frequency of distressing dreams qualifying as a nightmare, per the DSM-5 nightmare disorder definition was at least once a week (c.f. Richards et al. 2023). The broader study's inclusion criteria also included individuals with frequent episodes of dream enactment (based on the PSQI-A). However, all participants whose data contributed to the current analyses satisfied the nightmare inclusion criterion. Exclusion criteria included active psychosis or severe cognitive impairment associated with a neurodegenerative disease (e.g., Alzheimer's disease) because of the barriers related to self-reporting of sleep patterns on the mobile app and administering sleep equipment (for sleep psychological measurement in the parent study; Richards et al. 2025). See Table 1 for participant demographics and clinical characteristics.

2.2 | Procedures

The Institutional Review Board for the University of California, San Francisco and San Francisco Veterans Affairs Health Care System approved these study procedures. All study procedures were conducted remotely, meaning that all measurement devices were mailed to participants' homes and all training and hook-up procedures were performed by study staff through live videoconferencing. Clinical interviews were conducted via telephone and self-report surveys were completed using online surveys during scheduled phone appointments with study staff. Enrolled participants completed 21 days of sleep diary entries using a mobile app (Richards 2018). Diary entries were completed in the morning and evening. The morning diary asked participants to rate their mood upon awakening, the number of distressing dreams they experienced, and subjective sleep quality for the prior night's sleep period. Estimates of total sleep time (TST) and sleep efficiency were calculated based on standard sleep diary variables (bedtime, lights-out time, sleep latency, wake after sleep onset, final wake-up time and out-of-bed time; see Table S1). Evening assessments asked participants about their mood at bedtime as well as other information not analysed here (Table S1).

2.2.1 | Clinician-Administered PTSD Scale for DSM-5 (CAPS-5)

Participants completed the Clinician-Administered PTSD Scale for the DSM-5 (CAPS-5; Weathers et al. 2013), a 30-item structured interview that assesses PTSD symptom severity and diagnostic status as determined by the DSM-5. The questions on the CAPS-5 refer to a criterion A event identified by the participant (i.e., index trauma). The items are rated with a single severity score that incorporates both frequency and intensity of PTSD-related symptoms. Scores on the CAPS-5 range from 0 to 80, with higher scores indicating greater PTSD symptom

TABLE 1 | Participant demographics and clinical characteristics (N = 103).

Variable	n (%)	Mean (SD)
PTSD status		
PTSD–	38 (36.9)	
PTSD+	61 (59.2)	
Unknown	4 (3.9)	
CAPS-5		26.88 (11.24)
DERS total		89.30 (24.13)
SE		76.2 (11.4)
TST (min)		403.17 (62.55)
Sleep quality		51.2 (13.0)
Weekly average # of distressing dreams		4.1 (3.0)
Age (years)		48.5 (14.7)
Sex		
Female	31 (30.1)	
Male	72 (69.9)	
Race		
White	66 (64.1)	
Black/African American	11 (10.7)	
Asian	5 (4.9)	
American Indian/Alaska Native	5 (4.9)	
Mixed	14 (13.6)	
Other	2 (1.9)	
Ethnicity		
Not Hispanic or Latino/a	91 (88.3)	
Hispanic or Latino/a	12 (11.7)	
Veteran status		
Non-veteran	13 (12.6)	
Veteran	90 (87.4)	

Abbreviations: CAPS-5 = Clinician-Administered PTSD Scale for DSM-5; DERS = Difficulties with Emotion Regulation Scale; PTSD = posttraumatic stress disorder; SE = sleep efficiency based on sleep diary; TST = total sleep time based on sleep diary.

severity. The internal reliability of the CAPS-5 was acceptable (Cronbach's $\alpha = 0.88$). In the current study, the CAPS-5 was administered by a Masters-level clinician over the phone and typically prior to the onset of diary data collection.

2.2.2 | Difficulties With Emotion Regulation Scale (DERS)

Participants completed the Difficulties with Emotion Regulation Scale (DERS; Gratz and Roemer 2004), a 36-item self-report

measure that assesses six aspects of emotion regulation: (1) non-acceptance of emotional responses, (2) difficulty engaging in goal-directed behaviour, (3) impulse control difficulties, (4) lack of emotional awareness, (5) limited access to emotion regulation strategies and (6) lack of emotional clarity. The DERS asks respondents to indicate how frequently they experience emotion regulation difficulties and answer choices reflect their typical experience with each statement. Therefore, the DERS was used to measure trait emotion regulation, similar to an approach by Daros and Ruocco (2021) who operationalised the DERS total score as trait symptom dimension of emotion regulation underlying psychopathology. Items are rated on a scale of 1 = *almost never* to 5 = *almost always*. The impulsivity and clarity subscales scores range from 5 to 25, while the nonacceptance, awareness and strategies subscale scores range from 6 to 30. The total score of the DERS is calculated by summing all of the items, with scores ranging from 36 to 180, where higher scores indicate more difficulty regulating emotions. The DERS total exhibited acceptable internal reliability (Cronbach's $\alpha = 0.94$) (Schrepp 2020) as did each subscale (non-acceptance $\alpha = 0.90$, goals $\alpha = 0.87$, impulse $\alpha = 0.91$, awareness $\alpha = 0.85$, strategies $\alpha = 0.90$, clarity $\alpha = 0.84$). This measure was administered on the same day as the CAPS-5.

2.2.3 | Diary Mood Assessment

Participants rated their mood upon awakening (morning mood) and before bedtime (evening mood) using a Likert-type scale ranging from –3 (extremely bad) to 3 (extremely good). The instructions were as follows: 'Please rate your overall mood at the time of your final awakening [alternatively, 'at bedtime', for bedtime diary] on a scale from –3 (*extremely bad*) to +3 (*extremely good*). A 'bad' mood refers to having sad, anxious, angry, or other 'negative' feelings. A 'good' mood refers to having happy, joyful, or other 'positive' feelings.

2.2.4 | Diary Distressing Dream Assessment

Each morning, participants reported how many distressing dreams they had during the prior night in response to the question: 'How many distressing dreams did you have last night?' Participants were provided with the following guidance for this item during sleep diary orientation: 'A distressing dream is a dream in which unpleasant events happen and in which you feel unpleasant emotions, most often fear and anxiety. You may not always remember the exact details of the dream when you wake up or later on, but you should at least recall some brief images or events. Having a distressing dream is different from waking up frightened or upset but with no memory of any dream content. Enter how many distressing dreams you had last night. Answer '0' if you cannot recall any dreams or if all dreams you had were pleasant or completely neutral.' Participants could access this description each time they opened this item by clicking on the associated 'information' icon. For the purposes of this analysis, we compute a distressing dream indicator taking a value of '0' for no distressing dreams and '1' for one or more distressing dreams. We note that participants could report up to 10 distressing dreams per night, but reports of two or more distressing dreams were relatively rare (12.1% of morning reports). We define *distressing dream burden* to be the proportion of nights on which participants had at least one distressing dream.

2.3 | Statistical Methods

All statistical analyses were performed in R version 4.3.3 (R Core Team 2024). Random-intercept cross-lag panel models (RI-CLPM) (Hamaker et al. 2015; Mulder and Hamaker 2021) were fit using the *lavaan* package (Rosseel 2012) and used to model (1) the bidirectional relationship between mood and distressing dreams (within-participant effects) and (2) the association between difficulties with emotion regulation (DERS) (and its subscales) and distressing dream burden, average mood and average SE (between-participants effects) (Hamaker et al. 2015; Mulder and Hamaker 2021).

RI-CLPMs are an extension of the traditional CLPM and account for ‘stable, trait-like differences between [subjects]’ (Mulder and Hamaker 2021), such as biological sex at birth, when evaluating how lagged values of one variable are predictive of subsequent values of the same variable and one or more different variables. The models decompose outcomes of interest into grand means across the sample, stable between-participant means (random intercepts) and fluctuating, within-participant components (Mulder and Hamaker 2021). In the context of this study, we decomposed mood reports into overall average mood, average mood for each participant (B_{Mood_i}), and within-participant fluctuations in mood over time. Similarly, we decomposed distressing dream reports into a grand average distressing dream burden, distressing dream burden for each participant (B_{DD_i}), and within-participant fluctuations in distressing dream burden from 1 day to the next. The same approach was adopted for sleep efficiency (B_{SE_i}), which was included in all models as a time-varying confound.

We used these models to test the following hypotheses: (1) mood in the morning predicts mood in the evening (H1a), which in turn (2) predicts the risk for having one or more distressing dreams (H1b). We hypothesised that distressing dreams would then predict mood on the following morning (H1c), leading to feedback loops between worsening mood states and increased risk for distressing dreams. The RI-CLPM enables modelling how within-participant fluctuations in mood predict fluctuations in distressing dream experience while accounting for time-invariant confounds over the course of our study. Importantly, the within-participant components of such models can adjust for time-varying confounds such as sleep efficiency (TST/(bedtime to lights out time + lights out to wake time + wake time to out of bed time)*100). Such adjustment reduces the possibility that mood effects are simply related to objective sleep quality and disturbances separate from distressing dreams.

We were not only interested in modelling the within-participant fluctuations in mood and distressing dream experience but also in how between-subject differences in emotional regulation difficulties predicted differences in these experiences. Accordingly, we tested the hypotheses that (1) distressing dream burden and negative mood would be correlated (H2a) and that (2) difficulties with emotion regulation (DERS) at baseline would predict both distressing dream burden and mood (H2b). As exploratory analyses, we investigated which specific subscales of DERS were most predictive

of distressing dream burden and overall mood. Critically, between-subjects analyses do not account for time-invariant confounds; therefore, we adjust for biological sex at birth and age at the beginning of the study for H2a and H2b. Lastly, we tested the hypotheses that, relative to participants with good emotion regulation skills, participants with poor emotion regulation skills had an increased risk of experiencing a distressing dream following poor mood in the evening (H3a) and that the morning mood was worse following a distressing dream for participants with high difficulties with emotion regulation (H3b).

2.3.1 | Model Specification

RI-CLPMs were fit by extending the accompanying code from Mulder and Hamaker (2021) found at (<https://jeroendmulder.github.io/RI-CLPM/lavaan.html>). Our models closely correspond to Extension 1 (Mulder and Hamaker 2021) but differed in that mood and distressing dream reports did not take place at the same point in time (‘wave’) and therefore associations between mood and distressing dream fluctuations were modelled as regressions and not as covariances (Figure 1). The within-participant component of the model included paths from morning to evening mood, from evening to the subsequent morning mood, from one morning to the next, and from one evening to the next (these are first- and second-order autoregressive effects). We additionally included paths from the distressing dream indicator and sleep efficiency to the next morning and evening mood. Lastly, we included paths from both morning and evening mood onto the distressing dream indicator for the following night as well as a path from evening mood onto sleep efficiency. Residual variances were allowed to covary between the distressing dream indicator and sleep efficiency.

Grand means of distressing dream burden, sleep efficiency and mood were allowed to vary over time. Residual variances of each variable were constrained to be equal over time, as were the within-subject parameters of interest. The distressing dream indicator was treated as continuous due to convergence problems with models that treated it as categorical. We report below parameter estimates along with their standard errors, followed in parentheses ($b(se)$) in addition to standardised parameter estimates (b_{std}) to facilitate comparison of effect sizes among different paths.

We report the Tucker–Lewis index (TLI), comparative fit index (CFI), root mean square error of approximation (RMSEA) and standardised root mean square residual (SRMR) of all RI-CLPMs. We adopted the model-fit criteria proposed by Hu and Bentler (TLI > 0.95, CFI > 0.95, RMSEA < 0.06 and SRMR < 0.08) (Hu and Bentler 1999).

2.3.2 | Missing Data

Missing data were multiply imputed using the *Amelia* R package (Honaker et al. 2011). Imputation models included the following daily level predictors: day of the study (0% missing), morning mood (4.9% missing), evening mood (6.0% missing),

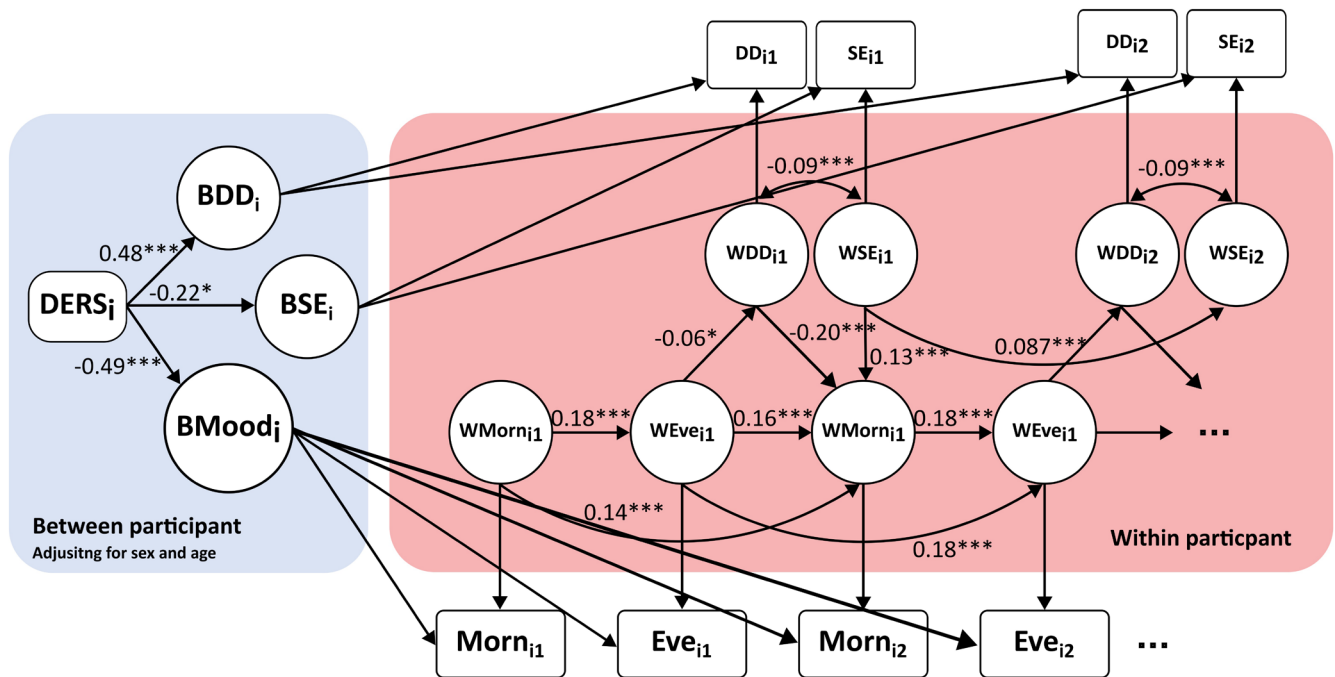


FIGURE 1 | Random intercept cross-lag panel model. Standardised values of the parameter estimates are shown. Nonsignificant paths are omitted from the figure. $DERS_i$ = difficulties with emotion regulation scale for subject i , BDD_i = distressing dream burden random intercept, $BMood_i$ = mood random intercept, SE_i = sleep efficiency random intercept. $WMorn_{it}$ = within-participant fluctuation from average mood in the morning for participant i on day t , $WEve_{it}$ = within-participant fluctuation in evening mood for participant i on day t , WDD_{it} = within-participant fluctuation in distressing dream experience for participant i on day t , WSE_{it} = within-participant fluctuation in sleep efficiency for participant i on day t . The three dots indicate that the model continues for 21 days. * $p < 0.05$, *** $p < 0.001$. This figure was adapted from Mulder and Hamaker (2021).

distressing dream indicator (5.1% missing), number of distressing dreams (5.1% missing), sleep quality (4.9% missing), sleep efficiency (6.2% missing) and total sleep time (6.2% missing). Lagged and leading values of each daily level predictor were also included. Depression (Patient Health Questionnaire-2; PHQ-2) (10.7% missing), PTSD severity (Posttraumatic Stress Disorder-8; PTSD-8) (10.4% missing) and suicide ideation (Beck Depression Inventory; BDI) (10.7% missing) scores at the end of each week of the study were additionally included, as were biological sex at birth (0% missing), each of the DERS subscales (5.8% missing), CAPS-5 total score (3.9% missing) and age (1.0% missing). Twenty simulated datasets were generated and models were fit using the *lavaan.mi()* function in the *semTools* package (Jorgensen et al. 2022) which fits the RI-CLPM model to each imputed dataset and pools the results using Rubin's rules (Rubin 1987).

3 | Results

Participants ($N = 103$, 90 veterans, 31 females) completed their morning diaries in a timely manner, with 64.6% of reports taking place on the morning they were assigned and 76.0% of evening reports filled out on the same evening they were assigned. Participants reported at least one distressing dream on 44.9% of nights. All models reported below demonstrated adequate fit with $TLI > 0.95$, $CFI > 0.95$, $RMSEA < 0.06$ and $SRMR > 0.08$ (Table S2).

3.1 | Within-Participant Effects

H1a: As depicted in Figure 1, mood carried over from morning to evening ($b = 0.18$ (0.023), $b_{std} = 0.18$, $t(462.48) = 7.68$, $p < 0.001$), from evening to morning ($b = 0.16$ (0.023), $b_{std} = 0.16$, $t(914.10) = 6.89$, $p < 0.001$), from one morning to the next ($b = 0.14$ (0.023), $b_{std} = 0.14$, $t(422.96) = 5.88$, $p < 0.001$), and from one evening to the next ($b = 0.18$ (0.024), $b_{std} = 0.18$, $t(574.54) = 7.39$, $p < 0.001$), independent of variation in sleep efficiency (Figure 1).

H1b: Again adjusting for sleep efficiency, mood in the evening was negatively correlated with the risk for having at least one distressing dream during the subsequent night ($b = -0.025$ (0.010), $b_{std} = -0.061$, $t(940.77) = -2.51$, $p = 0.012$) with more negative mood associating with higher risk. Morning mood was not predictive of having at least one distressing dream on the following night after accounting for evening mood ($b = 0.003$ (0.010), $b_{std} = 0.01$, $t(1143.89) = 0.31$, $p = 0.76$). We additionally found that fluctuations in sleep efficiency were not predicted by fluctuations in evening mood ($b = 0.044$ (0.031), $b_{std} = 0.03$, $t(6656.14) = 1.40$, $p = 0.16$).

H1c: Having at least one distressing dream was associated with a more negative mood in the morning ($b = -0.50$ (0.056), $b_{std} = -0.20$, $t(962.41) = -8.85$, $p < 0.001$) but not in the evening ($b = -0.010$ (0.057), $b_{std} = -0.004$, $t(1152.11) = -0.17$, $p = 0.86$) after accounting for morning mood and sleep efficiency.

We additionally evaluated the autocorrelation within distressing dream experiences and sleep efficiency. Here, we found that distressing dreams did not evidence significant lag 1 autocorrelation from one night to the next ($b=0.014$ (0.025), $b_{std}=0.014$, $t(2178.43)=0.58$, $p=0.56$) while sleep efficiency did ($b=0.087$ (0.024), $b_{std}=0.087$, $t(2901.40)=3.56$, $p<0.001$), meaning that sleep efficiency tended to carry over from one night to the next, whereas the likelihood of having at least one distressing dream did not carry over.

3.2 | Between-Participant Effects

H2a: Distressing dream burden was negatively correlated with negative mood over the duration of the study (Covariance = -0.049 (0.016), $r=-0.38$, $z=-2.96$, $p=0.003$), indicating that as distressing dream burden increased, average mood decreased (Table 2).

H2b: Adjusting for sex and age, difficulty with emotion regulation (DERS score) was positively associated with distressing dream burden ($b=0.005$ (0.001), $b_{std}=0.48$, $z=4.60$, $p<0.001$) and negatively associated with mood ($b=-0.016$ (0.003), $b_{std}=-0.49$, $z=-5.41$, $p<0.001$) and sleep efficiency ($b=-0.009$ (0.004), $b_{std}=-0.22$, $z=-2.03$, $p=0.043$) (Table 2).

Exploratory analyses of the DERS subscales revealed that distressing dream burden was positively associated with difficulties with nonacceptance ($b=0.15$ (0.004), $b_{std}=0.38$, $z=3.41$, $p=0.001$), goals ($b=0.020$ (0.005), $b_{std}=0.43$, $z=4.12$, $p<0.001$), impulse ($b=0.021$ (0.005), $b_{std}=0.47$, $z=4.61$, $p<0.001$) and strategies ($b=0.014$ (0.003), $b_{std}=0.43$, $z=4.05$, $p<0.001$) but not with difficulties in awareness ($b=0.006$ (0.005), $b_{std}=0.13$, $z=1.20$, $p=0.23$) or clarity ($b=0.010$ (0.007), $b_{std}=0.18$, $z=1.55$, $p=0.12$)

(Table 2). Average mood was negatively predicted by the nonacceptance ($b=-0.051$ (0.013), $b_{std}=-0.38$, $z=-3.85$, $p=0.001$), goals ($b=-0.061$ (0.015), $b_{std}=-0.37$, $z=-4.01$, $p<0.001$), impulse ($b=-0.07$ (0.014), $b_{std}=-0.46$, $z=-5.05$, $p<0.001$), strategies ($b=-0.047$ (0.010), $b_{std}=-0.42$, $z=-4.50$, $p<0.001$) and clarity ($b=-0.074$ (0.019), $b_{std}=-0.37$, $z=-3.81$, $p<0.001$) subscales but not by awareness ($b=-0.018$ (0.015), $b_{std}=-0.12$, $z=-1.17$, $p=0.24$).

We next included the CAPS-5 as a predictor of both average mood, distressing dream burden and sleep efficiency to adjust for the potential confounding effect of PTSD severity. This model revealed that the DERS remained significantly predictive of both distressing dream burden ($b=0.004$ (0.001), $b_{std}=0.39$, $z=3.23$, $p=0.001$) and average mood ($b=-0.012$ (0.003), $b_{std}=-0.35$, $z=-3.36$, $p=0.001$) even when accounting for PTSD severity (Table 2). The CAPS-5 was predictive of overall mood ($b=-0.019$ (0.007), $b_{std}=-0.27$, $z=-2.57$, $p=0.01$) but, surprisingly, not predictive of distressing dream burden when adjusting for the DERS, sex and age ($b=0.003$ (0.003), $b_{std}=0.16$, $z=1.29$, $p=0.20$).

H3a: We next sought to test the hypothesis that evening mood would be a stronger predictor of risk for having a distressing dream in participants with poor emotion regulation skills relative to participants with good emotion regulation skills. That is, we were interested in testing the hypothesis that the association between evening mood and risk for having a distressing dream was moderated by the DERS and its subscales. To do so, we fit the RI-CLPM separately for low and high DERS participants (determined by median split) while allowing the path between evening mood and the distressing dream indicator to differ between low and high DERS participants (determined by median split) and by constraining all other parameters in the model to

TABLE 2 | Between-participant emotion regulation predictors of distressing dream burden and average mood.

Predictor	Distressing dream burden				Average mood			
	Beta (se)	Standardised beta	Z	p	Beta (se)	Standardised beta	Z	p
DERS total	0.005 (0.001)	0.48	4.60	<0.001	−0.016 (0.003)	−0.49	−5.41	<0.001
DERS total (CAPS-5 adjusted)	0.004 (0.001)	0.39	3.23	0.001	−0.012 (0.003)	−0.35	−3.36	0.001
Nonacceptance	0.015 (0.004)	0.38	3.41	0.001	−0.051 (0.013)	−0.38	−3.85	<0.001
Goals	0.020 (0.005)	0.43	4.12	<0.001	−0.061 (0.015)	−0.37	−4.01	<0.001
Impulse	0.021 (0.005)	0.47	4.61	<0.001	−0.07 (0.014)	−0.46	−5.05	<0.001
Awareness	0.006 (0.005)	0.13	1.20	0.23	−0.018 (0.015)	−0.12	−1.17	0.24
Strategies	0.014 (0.003)	0.43	4.05	<0.001	−0.047 (0.010)	−0.42	−4.50	<0.001
Clarity	0.010 (0.007)	0.18	1.55	0.12	−0.074 (0.019)	−0.37	−3.81	<0.001

Note: All models adjusted for sex and age.

Abbreviations: CAPS-5 = Clinician-Administered PTSD Scale for DSM-5; Clarity = Lack of emotional clarity is a subscale of the DERS that includes 5 items pertaining to the extent to which an individual knows and is clear about their emotions; DERS = Difficulties with emotion regulation scale; Goals = Difficulty engaging in goal-directed behaviour is a subscale of the DERS that includes 5 items pertaining to difficulty in concentrating and/or accomplishing tasks when experiencing negative emotions; Impulse = Impulse control difficulties is a subscale of the DERS that includes 6 items pertaining to difficulty remaining in control of one's behaviour when experiencing negative emotions; Nonacceptance = Nonacceptance of emotional responses is a subscale of the DERS that includes 5 items pertaining to a tendency to have negative secondary or non-accepting reactions to one's own distress; Strategies = Limited access to emotion regulation strategies is a subscale of the DERS that includes 8 items pertaining to beliefs that there is little one can do to regulate oneself once upset.

be equal across DERS groups. These analyses were based on data from the final week of data collection as there were insufficient degrees of freedom to fit a model on the complete dataset. Contrary to our hypothesis, the association between evening mood and risk for distressing dreams did not differ between high and low DERS participants ($F(1, 441.2)=0.0, p=1.0$). Further, none of the DERS subscales moderated the association between evening mood and risk for having at least one distressing dream ($ps > 0.05$).

H3b: We lastly tested the hypothesis that the impact of distressing dreams on morning mood was moderated by DERS. Again, all model parameters were constrained to be equal across DERS groups apart from the path between the distressing dream indicator and next morning mood. Contrary to our hypothesis, DERS did not significantly moderate the effect distressing dreams had on the subsequent morning's mood ($F(1, 211.89)=0.39, p=0.53$). Similarly, no significant moderation effects were obtained for the DERS subscales ($ps > 0.05$).

In summary, participants with poor emotion regulation skills had more distressing dreams and worse mood on average, but their risk for having at least one distressing dream due to fluctuations in mood and momentary fluctuations in mood following distressing dreams did not differ from participants with good emotion regulation skills.

3.3 | Distress Severity

Given that previous research has shown that distressing dream distress, rather than frequency, is linked to poorer psychological outcomes upon awakening (Levin and Fireman 2002), additional exploratory analyses were performed to investigate whether DERS total and subscale scores predicted distressing dream severity. For these analyses, linear mixed effects models were used to examine the correlation between the DERS and distress severity conditional on having at least one distressing dream. Models adjusted for day of the study, age, biological sex at birth and sleep efficiency (both person-mean-centred and mean sleep efficiency for the person were included in the model to separate within- and between-subject effects, respectively).

Results of these analyses reveal the DERS total score was positively correlated with distress severity ($b=0.01$ (0.005), $t(87.32)=2.89, p=0.005$). Specifically, difficulties with goals ($b=0.07$ (0.02), $t(89.05)=3.15, p=0.002$), impulse ($b=0.07$ (0.02), $t(83.95)=3.52, p<0.001$) and strategies ($b=0.05$ (0.02), $t(88.98)=2.80, p=0.006$) were predictive of higher distress severity, whereas difficulties with nonacceptance ($b=0.03$ (0.02), $t(85.39)=1.53, p=0.13$), awareness ($b=0.00$ (0.02), $t(89.64)=0.02, p=0.98$) and clarity ($b=0.04$ (0.03), $t(91.69)=1.28, p=0.20$) were not.

We additionally explored the degree to which distress severity predicted morning and evening mood while adjusting for the DERS, day of the study, age, biological sex at birth and sleep efficiency (both person-mean-centred and person mean sleep efficiency were included as predictors). Again, these models incorporated person-mean-centred mood and person mean mood to estimate within- and between-subjects effects of

distress severity on morning and evening mood. Here, distress severity was negatively correlated with morning mood while adjusting for the previous evening's mood ($b=-0.20$ (0.02), $t(707.20)=-9.12, p<0.001$). That said, distress severity was not associated with subsequent evening mood when adjusting for morning mood ($b=0.05$ (0.03), $t(769.80)=1.71, p=0.09$).

4 | Discussion

While previous research has established that distressing dreams are a key feature of PTSD and are associated with poor mental health, more research is needed to identify the separable factors that contribute to the incidence of distressing dreams in persons with PTSD. In addition, the consequences of these experiences are not well understood, as most studies have utilised cross-sectional methods, limiting the interpretability of the findings. To our knowledge, this is the first study to evaluate the bidirectional nature of the relationship between mood and distressing dreams and to explore the potential contributions of difficulties with emotion regulation on distressing dream burden and overall mood among trauma-exposed civilians and veterans. Ultimately, we found that individuals with more negative mood prior to bedtime had an increased risk of having at least one distressing dream, which negatively impacted their mood the following morning. This poor mood state transferred into the evening, thus increasing the risk for another distressing dream.

After adjusting for sex, age and PTSD severity, distressing dream burden was positively associated with negative mood and negatively associated with emotion regulation. Nevertheless, at the within-subject level, difficulties with emotion regulation did not impact the relationship between an individual's negative mood during the day and the likelihood of having a distressing dream that night. Hence, our findings are only partially consistent with the findings of Levin and Nielsen (2009) that individuals with difficulties with emotion regulation may have a stronger response to an affective load relative to those that are better at emotion regulation.

Exploratory analyses revealed that all DERS subscales—apart from awareness—were predictive of negative mood. Similarly, with the exception of awareness and clarity, each subscale predicted distressing dream burden. These findings are in line with past studies that have linked several of the DERS subscales to clinical outcomes of interest (Akram et al. 2020; Schantz et al. 2024; Zhou et al. 2023). For example, Schantz et al. (2024) identified that strategies and nonacceptance subscales were strong predictors of anxiety symptoms. Zhou et al. (2023) provided evidence that the strategies subscale mediated the relationship between sleep disturbances and PTSD severity. Akram et al. (2020) report that strategies, non-acceptance, impulse and clarity subscales mediated the relationship between features of nightmares and subsequent psychotic experiences.

Our findings suggest that awareness and clarity seem to reflect introspective processes rather than explicit behavioural responses/strategies. This is not consistent with research that suggests that emotional awareness is essential for effective emotion regulation (Barrett et al. 2001) and that lower than average levels of emotional awareness have been found in patients

with PTSD (Frewen et al. 2008). The finding that awareness was not predictive of mood or distressing dream burden in our study may be due to the fact that this subscale has been found to lack internal consistency (Hallion et al. 2018). Indeed, Hallion et al. (2018) demonstrate that the best fitting factor structure of the DERS was bifactor and consisted of one general factor and uncorrelated specific factors that included clarity, goals, impulse, nonacceptance and strategies, but not awareness.

These findings should be interpreted with several limitations in mind. First, sleep diary data were collected over a relatively short time span of 3 weeks. Therefore, we needed to restrict our analyses investigating how DERS moderated the within-subject effects to the final week of the study due to insufficient degrees of freedom. A longer-term follow-up with more participants may have revealed significant effects in the within-subject findings of DERS. Additionally, while it is critical to understand the factors that contribute to distressing dreams among military populations because they are more acutely affected by trauma and associated negative mental health outcomes relative to civilians, our findings are not as generalisable to civilians because our sample was primarily comprised of veterans. Similarly, the preponderance of males (69.9%) and white (64.1%) participants in our sample also hinders the generalisability of these results. Finally, while mood was measured in a succinct and user-friendly way in the form of a sleep diary, the mood measurement was not validated. Our limitations notwithstanding, our findings demonstrate important implications of poor mood states as a precipitant of distressing dreams and vice versa. Future research should continue to explore additional factors that contribute to distressing dreams to inform or refine treatment conceptualisation and approaches for trauma-exposed individuals.

Moreover, from a treatment perspective, it is important to establish treatment priorities, which require engaging patients in how they choose to receive care for PTSD-related symptoms, including distressing dreams. In a study with veterans in outpatient ($n=216$) and residential ($n=812$) care for PTSD who were asked to provide 2–3 open-ended problems they hoped to improve during treatment, the most common issues were sleep problems (14.3%–27.3%) and nightmares (12.3%–19.4%) (Rosen et al. 2013). Therefore, it is important that future research continue to explore novel contributions to distressing dreams among trauma-exposed individuals.

Our findings provide additional rationale for the development of tools and interventions enabling trauma-exposed individuals to strengthen their emotion regulation abilities and engage in activities that improve daytime mood states. Future intervention research targeting emotion regulation, mood and/or distressing dreams in trauma survivors would benefit from similar analyses to examine how change in each of these factors influences the others.

Author Contributions

Nadia Malek: writing – original draft, writing – review and editing, data curation, conceptualization, formal analysis, project administration. **Anthony Santistevan:** formal analysis, data curation,

writing – review and editing, writing – original draft, conceptualization, methodology, visualization. **Leslie M. Yack:** project administration, conceptualization, investigation, funding acquisition, writing – review and editing. **Miles Kovnick:** project administration, writing – review and editing, data curation. **Shane Pracar:** project administration, writing – review and editing, data curation. **Emily Berg:** project administration, writing – review and editing, data curation, funding acquisition. **Thomas J. Metzler:** project administration, conceptualization, methodology, writing – review and editing, funding acquisition. **Steven H. Woodward:** conceptualization, funding acquisition, writing – review and editing, methodology, investigation. **Thomas C. Neylan:** project administration, conceptualization, funding acquisition, writing – review and editing, investigation. **Anne Richards:** supervision, resources, writing – review and editing, investigation, funding acquisition, conceptualization, writing – original draft, methodology, formal analysis, project administration, data curation.

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Ethics Statement

Approval for this study was granted by the Institutional Review Board at the University of California, San Francisco.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

De-identified data will be made available upon reasonable request to the corresponding author.

References

- Akram, U., M. Gardani, K. Irvine, et al. 2020. “Emotion Dysregulation Mediates the Relationship Between Nightmares and Psychotic Experiences: Results From a Student Population.” *NPJ Schizophrenia* 6, no. 1: 1–7. <https://doi.org/10.1038/s41537-020-0103-y>.
- American Psychiatric Association. 1994. *Diagnostic and Statistical Manual of Mental Disorders*. 4th ed. American Psychiatric Publishing.
- American Psychiatric Association. 2013. *Diagnostic and Statistical Manual of Mental Disorders*. 5th ed. American Psychiatric Publishing.
- Barrett, L. F., J. Gross, T. C. Christensen, and M. Benvenuto. 2001. “Knowing What You’re Feeling and Knowing What to Do About It: Mapping the Relation Between Emotion Differentiation and Emotion Regulation.” *Cognition & Emotion* 15, no. 6: 713–724.
- Blanchard, A. W., K. Rufino, and M. A. Patriquin. 2024. “Difficulties in Emotion Regulation Moderates the Relationship Between Mood Symptoms and Nightmares in an Inpatient Psychiatric Sample.” *Journal of Affective Disorders* 351: 179–183. <https://doi.org/10.1016/j.jad.2024.01.244>.
- Buyse, D. J., C. F. Reynolds III, T. H. Monk, S. R. Berman, and D. J. Kupfer. 1989. “The Pittsburgh Sleep Quality Index: A New Instrument for Psychiatric Practice and Research.” *Psychiatry Research* 28, no. 2: 193–213.
- Creamer, J. L., M. S. Brock, P. Matsangas, V. Motamedi, and V. Mysliwiec. 2018. “Nightmares in United States Military Personnel With Sleep Disturbances.” *Journal of Clinical Sleep Medicine* 14, no. 3: 419–426.

- Daros, A. R., and A. C. Ruocco. 2021. "Which Emotion Regulation Strategies Are Most Associated With Trait Emotion Dysregulation? A Transdiagnostic Examination." *Journal of Psychopathology and Behavioral Assessment* 43, no. 3: 478–490.
- Davis, J. L., K. E. Pruiksma, J. L. Rhudy, and P. Byrd. 2011. "A Comparison of Lifelong and Posttrauma Nightmares in a Civilian Trauma Sample: Nightmare Characteristics, Psychopathology, and Treatment Outcome." *Dreaming* 21, no. 1: 70–80.
- Frewen, P., R. D. Lane, R. W. Neufeld, M. Densmore, T. Stevens, and R. Lanius. 2008. "Neural Correlates of Levels of Emotional Awareness During Trauma Script-Imagery in Posttraumatic Stress Disorder." *Psychosomatic Medicine* 70, no. 1: 27–31.
- Germain, A., M. Hall, B. Krakow, M. K. Shear, and D. J. Buysse. 2005. "A Brief Sleep Scale for PTSD-Related Sleep Disturbances: The Pittsburgh Sleep Quality Index Addendum for PTSD." *Journal of Anxiety Disorders* 19, no. 2: 233–244. <https://doi.org/10.1016/j.janxdis.2004.02.001>.
- Gratz, K. L., and L. Roemer. 2004. "Multidimensional Assessment of Emotion Regulation and Dysregulation: Development, Factor Structure, and Initial Validation of the Difficulties in Emotion Regulation Scale." *Journal of Psychopathology and Behavioral Assessment* 26: 41–54.
- Hallion, L. S., S. A. Steinman, D. F. Tolin, and G. J. Diefenbach. 2018. "Psychometric Properties of the Difficulties in Emotion Regulation Scale (DERS) and Its Short Forms in Adults With Emotional Disorders." *Frontiers in Psychology* 9: 539. <https://doi.org/10.3389/fpsyg.2018.00539>.
- Hamaker, E. L., R. M. Kuiper, and R. P. P. Grasman. 2015. "A Critique of the Cross-Lagged Panel Model." *Psychological Methods* 20, no. 1: 102–116. <https://doi.org/10.1037/a0038889>.
- Honaker, J., G. King, and M. Blackwell. 2011. "Amelia II: A Program for Missing Data." *Journal of Statistical Software* 45, no. 7: 1–47. <https://doi.org/10.18637/jss.v045.i07>.
- Hooper, L., P. Stockton, J. Krupnick, and B. Green. 2011. "Development, Use, and Psychometric Properties of the Trauma History Questionnaire." *Journal of Loss and Trauma* 16: 258–283. <https://doi.org/10.1080/15325024.2011.572035>.
- Hu, L., and P. M. Bentler. 1999. "Cutoff Criteria for Fit Indexes in Covariance Structure Analysis: Conventional Criteria Versus New Alternatives." *Structural Equation Modeling: A Multidisciplinary Journal* 6, no. 1: 1–55. <https://doi.org/10.1080/10705519909540118>.
- Jorgensen, T. D., S. Pornprasertmanit, A. M. Schoemann, et al. 2022. "semTools: Useful Tools for Structural Equation Modeling (Version 0.5-6) [Computer Software]." <https://cran.r-project.org/web/packages/semTools/index.html>.
- Kessler, R. C., S. Heeringa, M. D. Lakoma, et al. 2008. "Individual and Societal Effects of Mental Disorders on Earnings in the United States: Results From the National Comorbidity Survey Replication." *American Journal of Psychiatry* 165, no. 6: 703–711. <https://doi.org/10.1176/appi.ajp.2008.08010126>.
- Köthe, M., and R. Pietrowsky. 2001. "Behavioral Effects of Nightmares and Their Correlations to Personality Patterns." *Dreaming* 11, no. 1: 43–52. <https://doi.org/10.1023/A:1009468517557>.
- Levin, R., and G. Fireman. 2002. "Nightmare Prevalence, Nightmare Distress, and Self-Reported Psychological Disturbance." *Sleep* 25, no. 2: 205–212.
- Levin, R., and T. Nielsen. 2009. "Nightmares, Bad Dreams, and Emotion Dysregulation: A Review and New Neurocognitive Model of Dreaming." *Current Directions in Psychological Science* 18, no. 2: 84–88. <https://doi.org/10.1111/j.1467-8721.2009.01614.x>.
- Milanak, M. E., K. L. Zuromski, I. Cero, A. K. Wilkerson, H. S. Resnick, and D. G. Kilpatrick. 2019. "Traumatic Event Exposure, Posttraumatic Stress Disorder, and Sleep Disturbances in a National Sample of US Adults." *Journal of Traumatic Stress* 32, no. 1: 14–22.
- Miller, K. E., E. M. Boland, H. Barilla, et al. 2024. "Ecological Momentary Assessment of Daily Affect, Stress, and Nightmare Reports Among Combat-Exposed Veterans." *Dreaming* 34: 307–317.
- Mulder, J. D., and E. L. Hamaker. 2021. "Three Extensions of the Random Intercept Cross-Lagged Panel Model." *Structural Equation Modeling: A Multidisciplinary Journal* 28, no. 4: 638–648. <https://doi.org/10.1080/10705511.2020.1784738>.
- Nadorff, M. R., S. Nazem, and A. Fiske. 2011. "Insomnia Symptoms, Nightmares, and Suicidal Ideation in a College Student Sample." *Sleep* 34, no. 1: 93–98. <https://doi.org/10.1093/sleep/34.1.93>.
- Nielsen, T., and R. Levin. 2007. "Nightmares: A New Neurocognitive Model." *Sleep Medicine Reviews* 11, no. 4: 295–310.
- Ohayon, M. M., P. L. Morselli, and C. Guilleminault. 1997. "Prevalence of Nightmares and Their Relationship to Psychopathology and Daytime Functioning in Insomnia Subjects." *Sleep* 20, no. 5: 340–348. <https://doi.org/10.1093/sleep/20.5.340>.
- R Core Team. 2024. *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing. <https://www.R-project.org/>.
- Richards, A. 2018. "The Sleep Diary App (Version 2.1.0) [Mobile Application Software]." <https://apps.apple.com/us/app/the-sleep-diary/id1477788812>.
- Richards, A., J. C. Kanady, and T. C. Neylan. 2020. "Sleep Disturbance in PTSD and Other Anxiety-Related Disorders: An Updated Review of Clinical Features, Physiological Characteristics, and Psychological and Neurobiological Mechanisms." *Neuropsychopharmacology* 45, no. 1: 55–73. <https://doi.org/10.1038/s41386-019-0486-5>.
- Richards, A., A. Santistevan, M. Kovnick, et al. 2025. "Distressing Dreams in Trauma Survivors: Using a Sleep Diary Mobile App to Reveal Distressing Dream Characteristics and Their Relationship to Symptoms and Suicidal Ideation in Trauma-Exposed Adults." *SLEEP Advances* 6: zpae099.
- Richards, A., S. H. Woodward, D. P. G. Baquirin, et al. 2023. "The Sleep Physiology of Nightmares in Veterans With Psychological Trauma: Evaluation of a Dominant Model Using Participant-Applied Electroencephalography in the Home Environment." *Journal of Sleep Research* 32, no. 2: e13639. <https://doi.org/10.1111/jsr.13639>.
- Richardson, L. K., B. C. Frueh, and R. Acierno. 2010. "Prevalence Estimates of Combat-Related Post-Traumatic Stress Disorder: Critical Review." *Australian and New Zealand Journal of Psychiatry* 44, no. 1: 4–19.
- Rosen, C., E. Adler, and Q. Tiet. 2013. "Presenting Concerns of Veterans Entering Treatment for Posttraumatic Stress Disorder." *Journal of Traumatic Stress* 26, no. 5: 640–643. <https://doi.org/10.1002/jts.21841>.
- Ross, R. J., W. A. Ball, K. A. Sullivan, and S. N. Caroff. 1989. "Sleep Disturbance as the Hallmark of Posttraumatic Stress Disorder." *American Journal of Psychiatry* 146, no. 6: 697–707. <https://doi.org/10.1176/ajp.146.6.697>.
- Rosseel, Y. 2012. "lavaan: An R Package for Structural Equation Modeling." *Journal of Statistical Software* 48, no. 2: 1–36. <https://doi.org/10.18637/jss.v048.i02>.
- Rubin, D. B. 1987. *Multiple Imputation for Nonresponse in Surveys*. J. Wiley & Sons. <https://doi.org/10.1002/9780470316696>.
- Rufino, K. A., E. F. Ward-Ciesielski, C. A. Webb, and M. R. Nadorff. 2020. "Emotion Regulation Difficulties Are Associated With Nightmares and Suicide Attempts in an Adult Psychiatric Inpatient Sample." *Psychiatry Research* 293: 113437. <https://doi.org/10.1016/j.psychres.2020.113437>.
- Schantz, B. L., E. R. Toner, M. L. Brown, et al. 2024. "Examining the Relationship Between Emotion Regulation, Sleep Quality, and Anxiety

Disorder Diagnosis." *Journal of Mood & Anxiety Disorders* 8: 100072. <https://doi.org/10.1016/j.xjmad.2024.100072>.

Schrepp, M. 2020. "On the Usage of Cronbach's Alpha to Measure Reliability of UX Scales." *Journal of Usability Studies* 15, no. 4: 247–258.

Seal, K. H., D. Bertenthal, and C. R. Miner. 2007. "Bringing the War Back Home: Mental Health Disorders Among 103 788 US Veterans Returning From Iraq and Afghanistan Seen at Department of Veterans Affairs Facilities." *Archives of Internal Medicine* 167: 476–482. <https://jamanetwork.com/journals/jamainternalmedicine/article-abstract/769661>.

Sheaves, B., S. Rek, and D. Freeman. 2023. "Nightmares and Psychiatric Symptoms: A Systematic Review of Longitudinal, Experimental, and Clinical Trial Studies." *Clinical Psychology Review* 100: 102241. <https://doi.org/10.1016/j.cpr.2022.102241>.

Sjöström, N., J. Hetta, and M. Waern. 2009. "Persistent Nightmares Are Associated With Repeat Suicide Attempt: A Prospective Study." *Psychiatry Research* 170, no. 2: 208–211. <https://doi.org/10.1016/j.psychres.2008.09.006>.

Tanielian, T. L., L. Jaycox, and Center for Military Health Policy Research. 2008. *Invisible Wounds of War: Psychological and Cognitive Injuries, Their Consequences, and Services to Assist Recovery*. Edited by T. L. Tanielian, L. Jaycox, and Center for Military Health Policy Research. Rand.

Weathers, F. W., D. D. Blake, P. P. Schnurr, D. G. Kaloupek, B. P. Marx, and T. M. Keane. 2013. *The Clinician-Administered PTSD Scale for DSM-5 (CAPS-5)*. [Assessment]. www.ptsd.va.gov.

Weber, F. C., and T. C. Wetter. 2022. "The Many Faces of Sleep Disorders in Post-Traumatic Stress Disorder: An Update on Clinical Features and Treatment." *Neuropsychobiology* 81, no. 2: 85–97. <https://doi.org/10.1159/000517329>.

Wisco, B. E., F. O. Nomamiukor, B. P. Marx, J. H. Krystal, S. M. Southwick, and R. H. Pietrzak. 2022. "Posttraumatic Stress Disorder in US Military Veterans: Results From the 2019–2020 National Health and Resilience in Veterans Study." *Journal of Clinical Psychiatry* 83, no. 2: 39779.

Worley, C. B., S. Meshberg-Cohen, I. C. Fischer, and R. H. Pietrzak. 2025. "Trauma-Related Nightmares Among US Veterans: Findings From a Nationally Representative Study." *Sleep Medicine* 126: 159–166.

Zhou, A., M. McDaniel, X. Hong, M. Mattin, X. Wang, and C.-H. Shih. 2023. "Emotion Dysregulation Mediates the Association Between Acute Sleep Disturbance and Later Posttraumatic Stress Symptoms in Trauma Exposed Adults." *European Journal of Psychotraumatology* 14, no. 2: 2202056. <https://doi.org/10.1080/20008066.2023.2202>.

Supporting Information

Additional supporting information can be found online in the Supporting Information section.